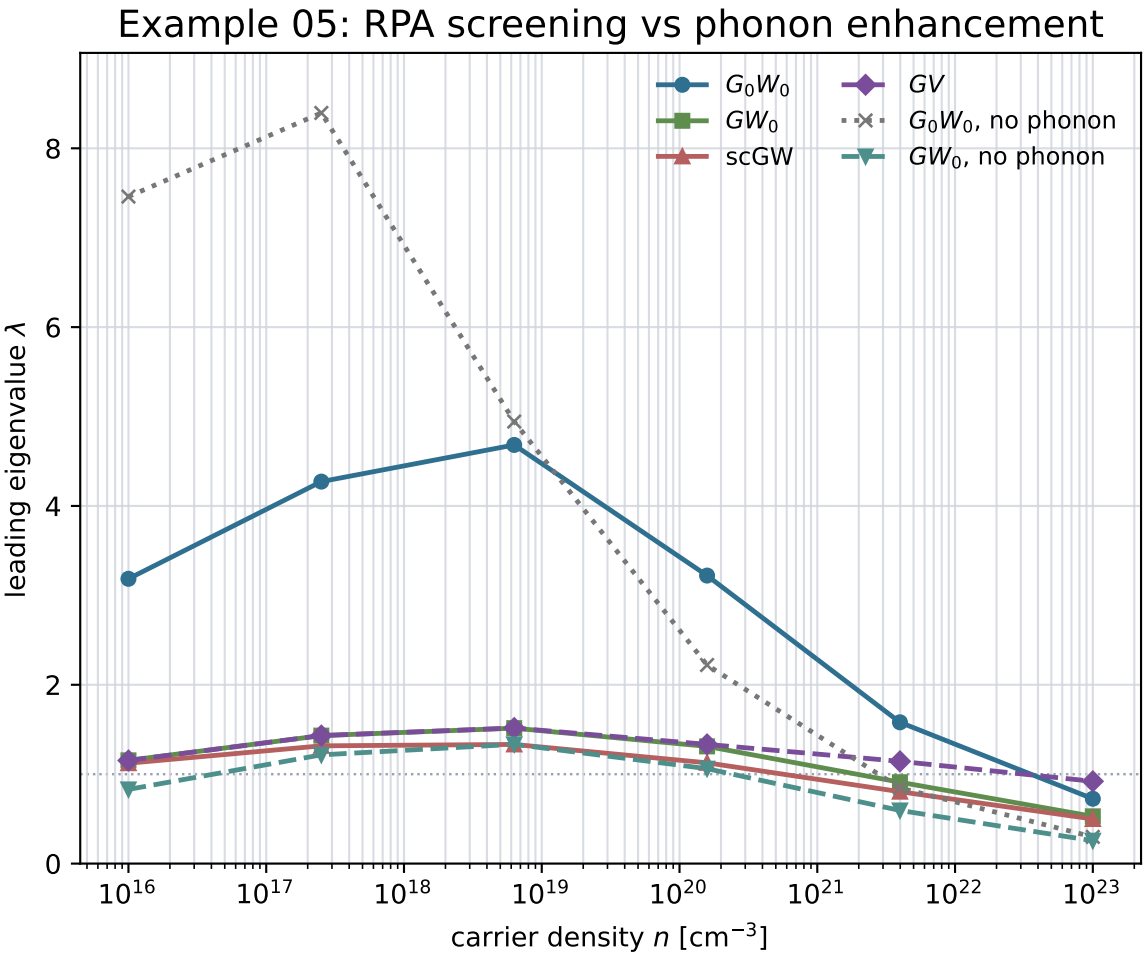

$W = V_{\mathrm{tot}}/(1 - \Pi V_{\mathrm{tot}})$, $V_{\mathrm{tot}} = V_{\mathrm{c}}\, V_{\mathrm{ph}}$; GV fixes $\Pi = 0$.
No-phonon sets $\varepsilon_0 = \varepsilon_\infty$, so $V_{\mathrm{ph}} = 1$ while electronic RPA screening remains.